

An Absolute-Space Derivation of Relativistic Time Dilation and the Relativistic Origin of Magnetism

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Abstract

This paper presents a first-principles derivation of special-relativistic time dilation, weak-field gravitational time dilation, and the magnetic force between moving charges, starting from a small set of classical postulates: an absolute three-dimensional space with a fixed, observer-independent distance structure; time treated not as an independent dimension but as a monotonic ordering derived from motion; and the assumption that every physical system possesses a perpetual internal process (an internal "signal") propagating at a fixed maximum rate. Combining this foundation with elementary geometry, Newtonian orbital mechanics, and the equivalence principle reproduces the standard Lorentz factor, the measured GPS relativistic clock corrections (approximately +45 microseconds per day from gravitational potential and -7 microseconds per day from orbital velocity), and the standard magnetic field of a current-carrying wire, all to good numerical agreement with established values. The paper is explicit about what this demonstrates and what it does not: every quantitative result obtained here is mathematically identical to the corresponding result in standard special and general relativity and classical electromagnetism. This work is presented as an alternative, internally consistent derivation of known physics from an absolute-space foundation, not as a new predictive theory. A concluding section identifies the specific gap - the absence of any point of numerical divergence from standard theory - that would need to be closed for this foundation to constitute a genuinely new, testable physical claim.

1. Foundational Postulates

The framework rests on four postulates, developed and pressure-tested against known physics over the course of this project:

Postulate 1 (Absolute Space). There exists a single, universal three-dimensional spatial structure with a fixed, observer-independent distance between any two points. No coordinate grid or reference frame is required for this structure to exist; the distance between two points is a real fact, not a convention.

Postulate 2 (Time as Ordering, Not Dimension). Time is not a dimension on equal footing with space. It is a monotonic, forward-only ordering of events, derived from the accumulation of motion rather than existing as an independent substance.

Postulate 3 (Eternal Motion). Motion is not caused; it is an eternal, foundational feature of the system. No physical account is required, or offered, for why motion exists in the first place.

Postulate 4 (Perpetual Internal Motion at a Fixed Rate). No physical system is ever truly at rest. Every particle possesses a perpetual internal process, modeled here as a signal propagating between two internal points, that travels at a fixed maximum rate, denoted c . Bulk motion of the system through absolute space is added on top of this internal motion, not in place of it.

Postulates 1 through 3 are philosophical commitments about the nature of space, time, and motion. Postulate 4 is the load-bearing physical assumption from which the quantitative results below are derived.

2. Special-Relativistic Time Dilation from Internal Motion

Consider a particle whose internal signal bounces between two fixed points separated by distance L , at fixed speed c .

At rest, one round trip takes:

$$T0 = 2L / c$$

When the particle moves at bulk velocity v through absolute space, the internal signal must trace a diagonal path to keep pace with the moving particle, since the two internal points have shifted sideways during the signal's transit. Applying the Pythagorean theorem to this diagonal path, with the signal still constrained to travel at fixed speed c , gives:

$$T = 2L / \sqrt{c^2 - v^2} = T0 / \sqrt{1 - v^2/c^2}$$

This is the standard Lorentz factor, $\gamma = 1/\sqrt{1 - v^2/c^2}$, obtained directly from Postulate 4 and elementary geometry. No target value was assumed in advance; the factor is a consequence of the fixed propagation rate of the internal signal combined with the geometry of bulk motion.

This derivation is mathematically identical to the standard "light clock" thought experiment used to introduce special relativity, and is closely related to the historical Lorentzian interpretation of relativity developed by Lorentz and Poincare, in which a real, physical absolute rest frame underlies relativistic effects. The Lorentzian and Einsteinian formulations are known to be empirically indistinguishable; this paper's derivation falls within the Lorentzian tradition.

3. Gravitational Time Dilation via the Equivalence Principle

Gravitational time dilation is obtained by combining the result of Section 2 with the equivalence principle: being at rest in a gravitational field of local strength g is physically indistinguishable from undergoing uniform acceleration $a = g$ through absolute space, in the absence of any mass.

Consider a light pulse emitted from the floor of a uniformly accelerating enclosure toward its ceiling, a height h above. The light's travel time is approximately $t = h/c$, during which the ceiling accumulates velocity $v = at = ah/c$ in the direction of the light's motion. The classical Doppler shift for this small velocity gives:

$$\Delta f/f \sim -v/c = -ah/c^2$$

Applying the equivalence principle ($a \rightarrow g$) and noting that gh is the gravitational potential difference ΔU over height h in a uniform field:

$$\Delta f/f \sim -\Delta U/c^2$$

Comparing clock rates directly rather than light in transit, a clock at higher gravitational potential runs faster than one at lower potential by:

$$\Delta f/f = +\Delta U/c^2$$

Generalizing from a uniform field to the true $1/r$ potential of a spherical mass, $\Delta U = GM(1/r_1 - 1/r_2)$, this becomes directly applicable to a satellite in orbit around a planet.

4. Numerical Verification Against GPS

The two results above were checked against the operational relativistic correction applied to GPS satellite clocks, using Earth's $GM = 3.986 \times 10^{14} \text{ m}^3/\text{s}^2$, Earth radius $6.371 \times 10^6 \text{ m}$, and GPS orbital radius $2.6571 \times 10^7 \text{ m}$ (orbital altitude $\sim 20,200 \text{ km}$).

Orbital velocity, from Newton's circular-orbit condition $GM/r^2 = v^2/r$, gives $v^2 = GM/r \sim 1.500 \times 10^7 \text{ m}^2/\text{s}^2$, or $v \sim 3.87 \text{ km/s}$, matching the known orbital speed of GPS satellites.

Velocity time dilation (Section 2): $\Delta f/f = -v^2/(2c^2) \sim -8.33 \times 10^{-11}$, equivalent to approximately -7.2 microseconds per day.

Gravitational time dilation (Section 3): using $\Delta U = GM(1/r_{\text{earth}} - 1/r_{\text{sat}}) \approx 4.759 \times 10^7 \text{ m}^2/\text{s}^2$, $\Delta f/f = \Delta U/c^2 \approx 5.29 \times 10^{-10}$, equivalent to approximately +45.7 microseconds per day.

Net predicted offset: approximately +38.5 microseconds per day, in close agreement with the well-established, operationally applied GPS relativistic correction of +38 microseconds per day (Ashby & Weiss; Fliegel & DiEposti).

5. Magnetism as a Relativistic Effect of Electric Charge

The framework was extended to electromagnetism by applying the same length-contraction consequence of Section 2 to a current-carrying wire, following the approach originally presented by Purcell (Electricity and Magnetism).

A current-carrying wire contains a lattice of positive charges at rest and a population of negative charges (electrons) drifting at velocity u ; the wire is electrically neutral in its own rest frame. Viewed from the rest frame of an external test charge moving at velocity v parallel to the wire, the two charge populations are moving at different velocities relative to the test charge, and therefore undergo different degrees of length contraction. This breaks the exact neutrality observed in the wire's own frame, producing a net charge density and therefore a real electric field in the test charge's frame, which exerts an ordinary electric force on the test charge.

Transforming this force back into the wire's rest frame reproduces exactly the standard magnetic force law, $F = qv \times B$, with B given by the standard formula for the field around a current-carrying wire, $B = \mu_0 I / (2\pi r)$.

A numerical check using a 1 A current in a copper wire (free electron density $\approx 8.5 \times 10^{28} \text{ m}^{-3}$, cross-sectional area 1 mm^2) gives an electron drift velocity of only $u \approx 7.35 \times 10^{-5} \text{ m/s}$, and a corresponding fractional charge imbalance between the two populations of order 3×10^{-26} , an extraordinarily small effect. That this vanishingly small relativistic imbalance, combined with the large prefactor of Coulomb's constant, reproduces the full, everyday strength of magnetic forces is the substantive content of the result: magnetism requires no independent fundamental field in this framework. It is fully accounted for as a relativistic consequence of electric charge and motion, using the same mechanism (Section 2) already responsible for special-relativistic time dilation.

6. Status and Open Questions

The results in Sections 2 through 5 are internally consistent, derived without free parameters chosen after the fact, and numerically verified against known values. The following points are stated plainly so that the scope of this work is not overstated.

6.1 - Every quantitative result in this paper is mathematically identical to the corresponding result in standard special relativity, general relativity (weak-field limit), and classical electromagnetism. No experiment currently known could distinguish this framework's predictions from those of standard theory, because the same underlying invariance (a fixed maximum signal speed, and the equivalence principle) was used in both derivations.

6.2 - This work is best understood as belonging to the Lorentzian tradition of interpreting relativity: it proposes a physically real absolute space and a mechanical account of why moving and gravitationally displaced clocks run at different rates, as an alternative to treating spacetime curvature as fundamental. It does not currently offer, and does not claim to offer, a new prediction that would empirically favor it over the standard formulation.

6.3 - Several candidate mechanisms explored during the development of this framework, including proposals that gravitational and magnetic field strength are generated directly by a body's rotation rate or core temperature, were tested against real planetary data (comparative surface gravity and magnetic field strength of Earth, Venus, and Jupiter) and did not hold up. These mechanisms have been abandoned and are not part of the framework presented here.

6.4 - The central open problem for this framework is identifying a specific physical regime in which it postulates force a numerical prediction different from standard relativity and electromagnetism. Absent such a divergence, this framework should be regarded as an alternative derivation and interpretation of established physics, not as a competing physical theory.

References

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